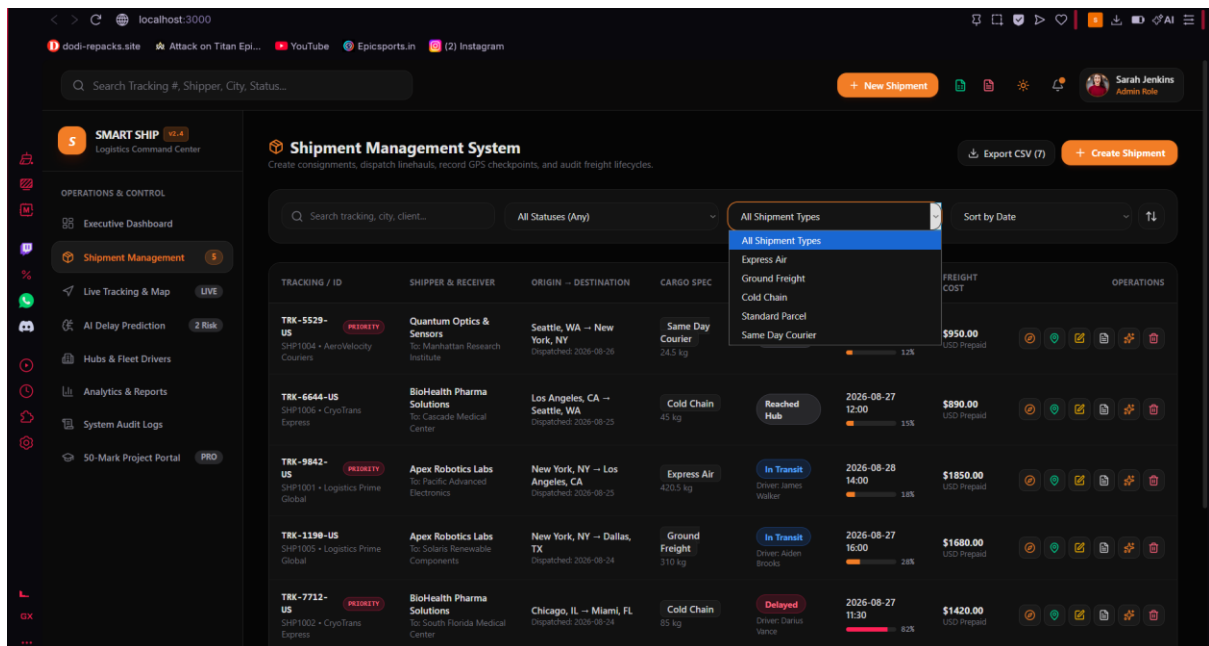
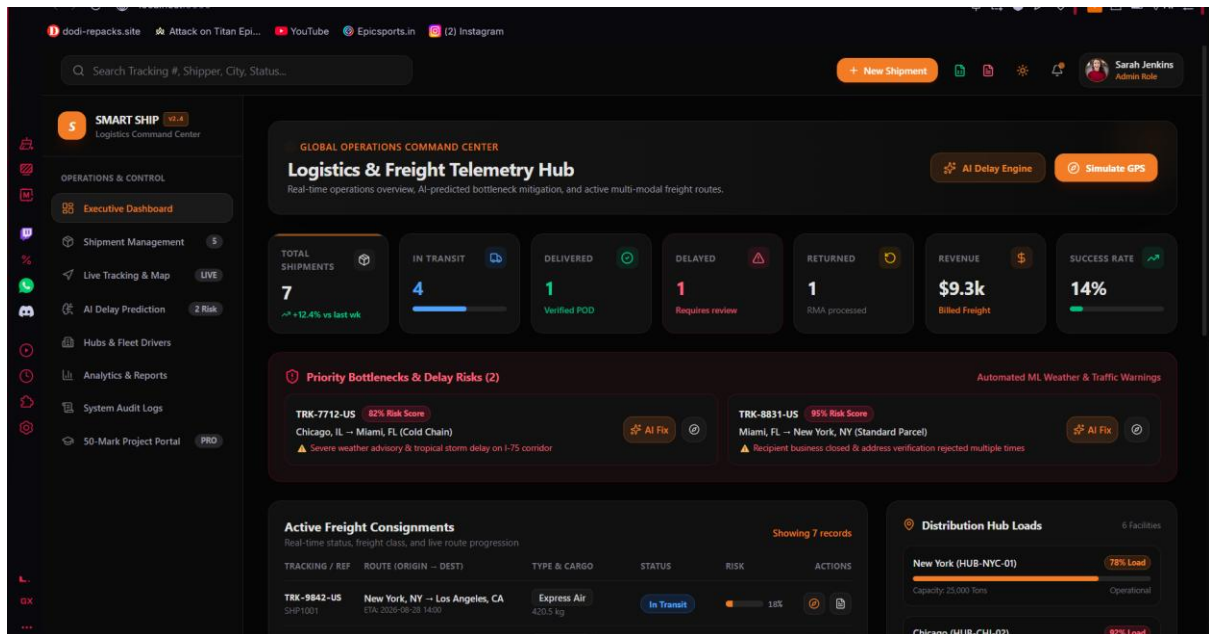
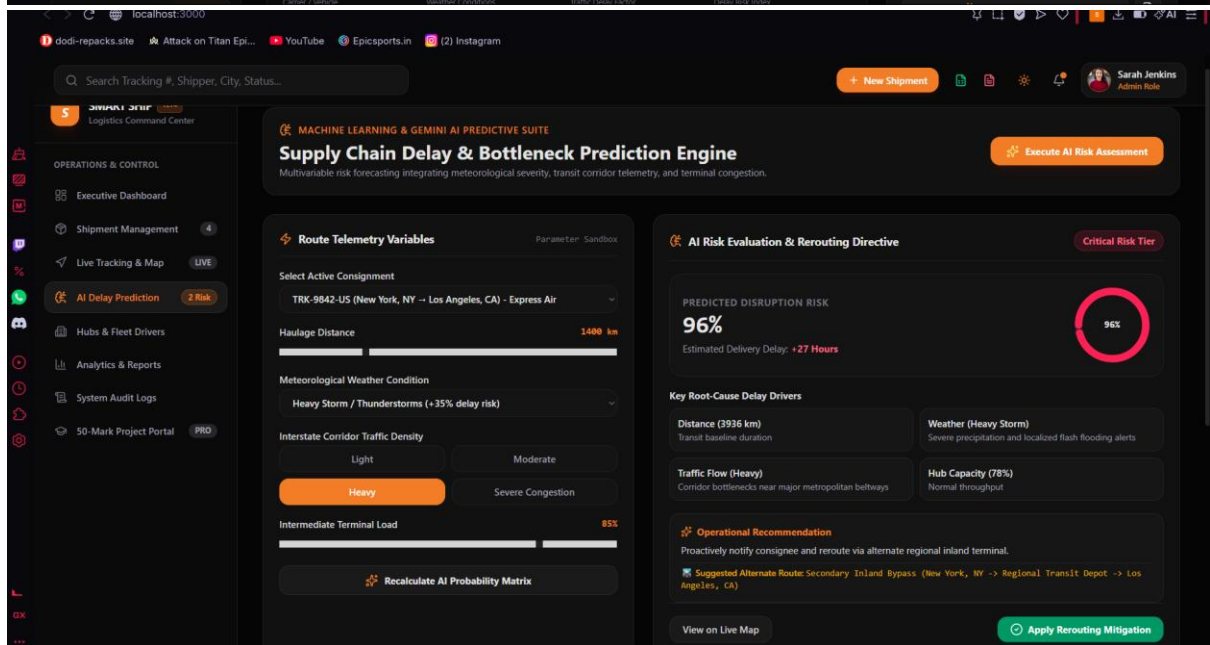
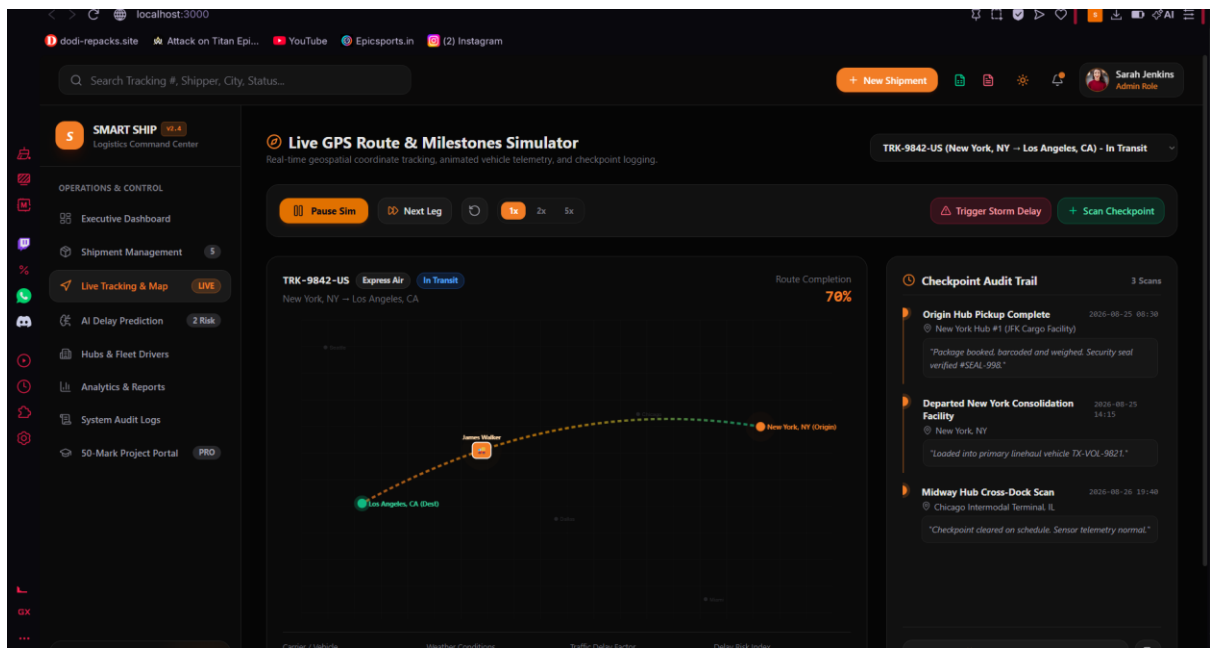
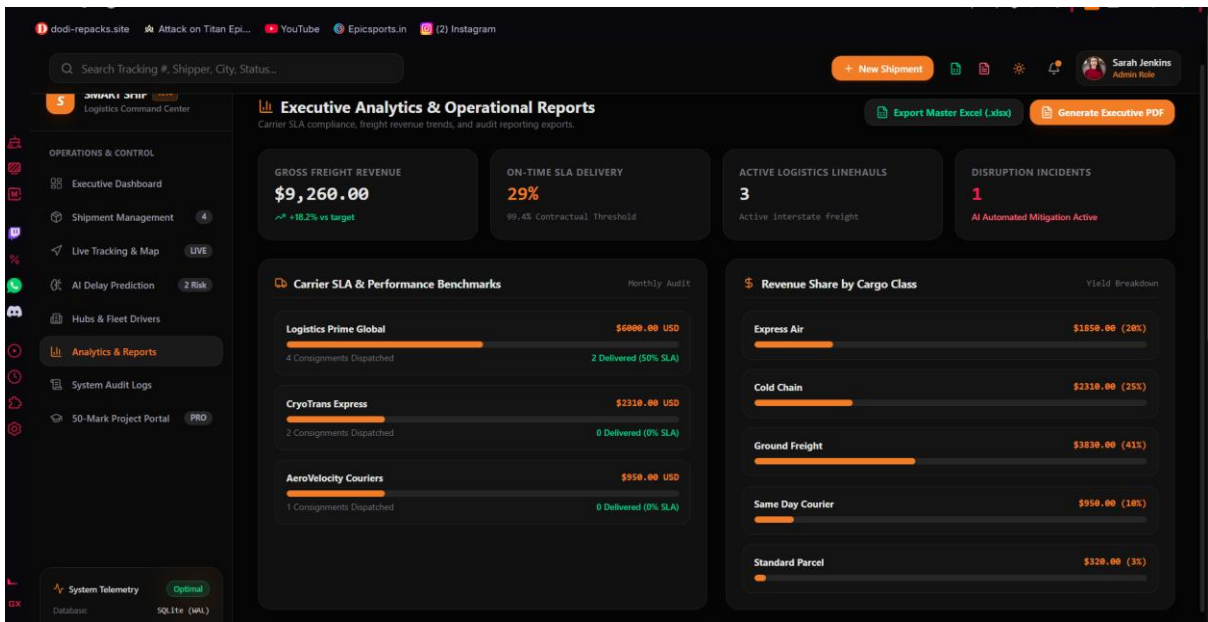
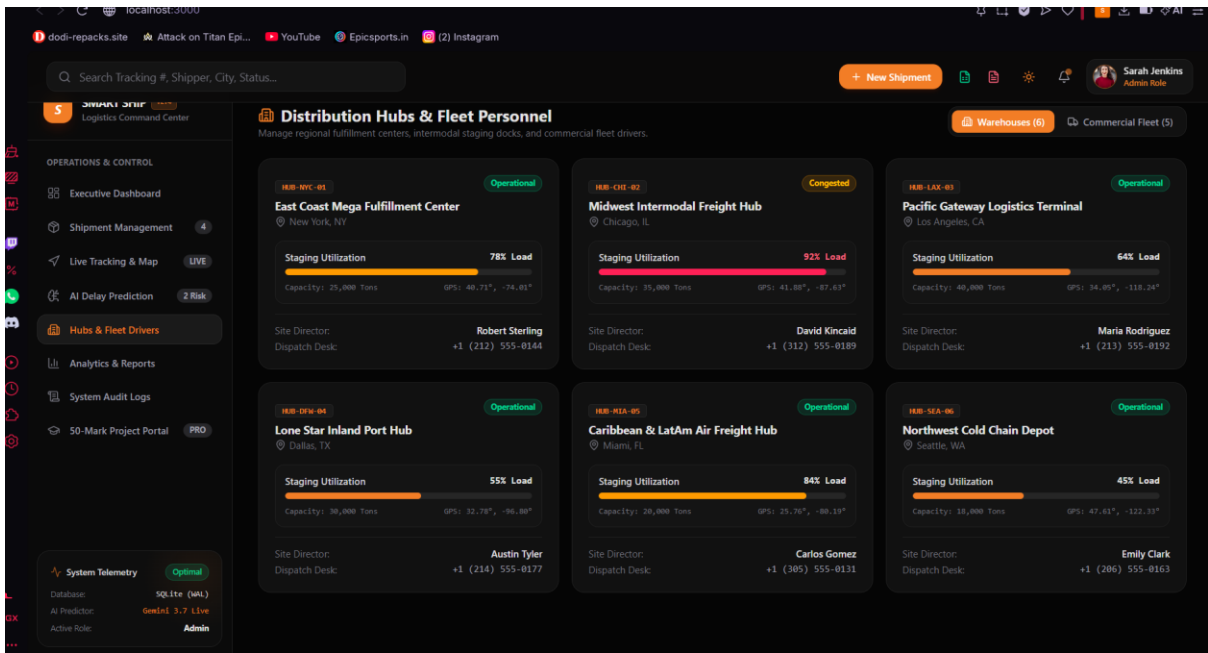
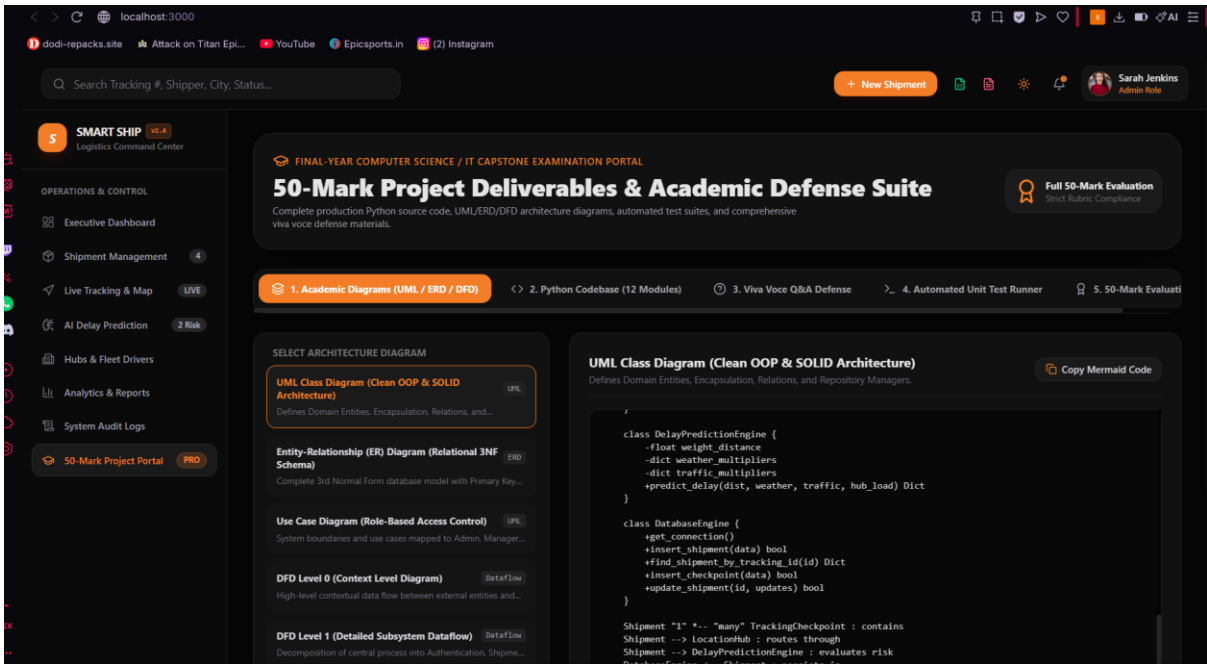
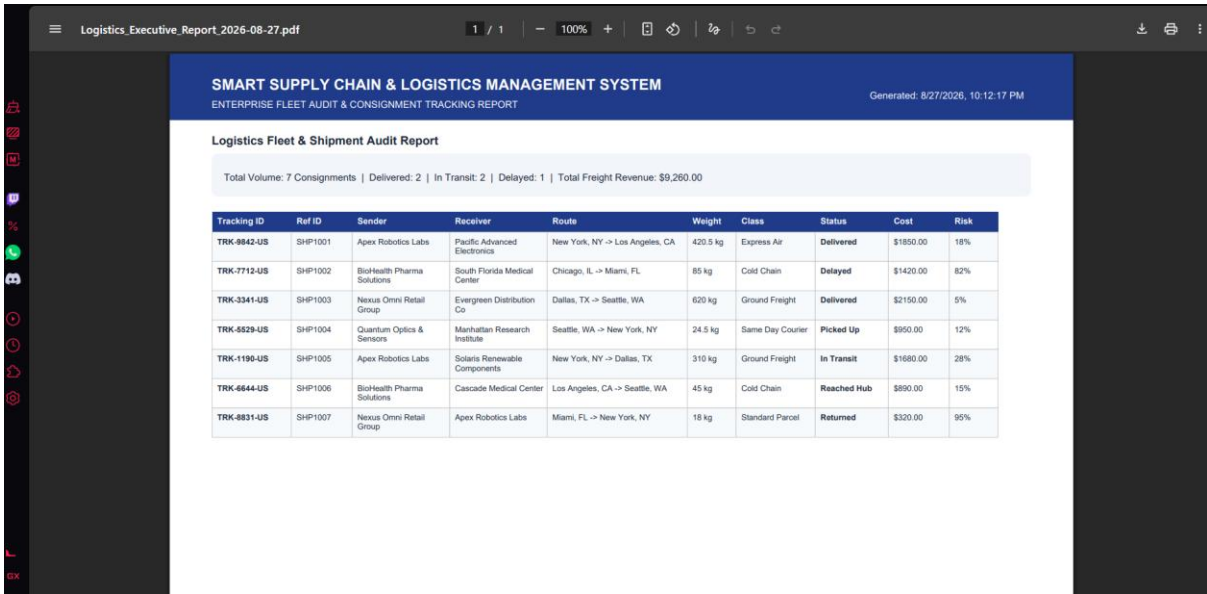


PROJECT URL :<http://localhost:3000>









classDiagram

```
class User {  
    +String id  
    +String username  
    +String password_hash  
    +String name  
    +UserRole role  
    +String email  
    +login(username, password) bool  
    +logout() void  
}
```

```
class Shipment {  
    +String tracking_id  
    +String shipment_id  
    +String sender_name  
    +String receiver_name  
    +String source  
    +String destination  
    +float weight  
    +String dimensions  
    +ShipmentType shipment_type  
    +DateTime shipment_date  
    +DateTime expected_delivery_date  
    +ShipmentStatus status  
    +float cost  
    +int delay_risk  
    +validate() bool
```

```
+calculate_freight_cost() float  
+update_status(new_status) void  
}
```

```
class TrackingCheckpoint {  
    +int id  
    +String tracking_id  
    +String checkpoint_name  
    +String location  
    +float latitude  
    +float longitude  
    +DateTime timestamp  
    +ShipmentStatus current_status  
    +String remarks  
    +String staff_name  
}
```

```
class LocationHub {  
    +String location_id  
    +String city  
    +String warehouse_name  
    +float latitude  
    +float longitude  
    +int capacity_tons  
    +int current_load_percent  
    +String status  
}
```

```

class DelayPredictionEngine {
    -float weight_distance
    -dict weather_multipliers
    -dict traffic_multipliers
    +predict_delay(dist, weather, traffic, hub_load) Dict
}

```

```

class DatabaseEngine {
    +get_connection()
    +insert_shipment(data) bool
    +find_shipment_by_tracking_id(id) Dict
    +insert_checkpoint(data) bool
    +update_shipment(id, updates) bool
}

```

Shipment "1" *-- "many" TrackingCheckpoint : contains

Shipment --> LocationHub : routes through

Shipment --> DelayPredictionEngine : evaluates risk

DatabaseEngine <-- Shipment : persists in

User "1" --> "many" Shipment : manages